JavaScript Problem Solving

Set 2: Problems 11 to 20, with hints

Is PDF ka istemal kaise karein

Har problem ko in **7 steps** mein hal karo. Seedha code mat likho, pehle copy par logic banao:

1. Problem, Input, Output copy par apne alfaaz mein likho.

2. Ek chhota example khud kagaz par hal karo (insaan ki tarah).

3. Algorithm likho: numbered steps, Start se End tak.

4. Pseudocode likho: SET, FOR, IF, RETURN jaise words ke saath.

5. Dry run karo: table banao (i, current value, result).

6. Edge cases socho: khali input, ek element, negative numbers.

7. Ab JavaScript mein code likho aur test cases chalao.

Is set mein naya kya hai

- Nested loop (loop ke andar loop): Problem 20.
- Prime check aur Fibonacci: classic problems jo interviews mein bhi aate hain.
- Ek loop mein do cheezein track karna (min aur max): Problem 17.
- Strings ko character level par samajhna: Problems 13, 14, 15.

Rules

- Pehle khud 20-30 minute koshish karo. Phir Hint 1, atko to Hint 2, phir Hint 3. Hints ek saath mat padhna.
- Jo built-in method (Math.max, reduce, includes ...) constraint mein mana hai, use mat karo. Maqsad logic seekhna hai.
- Function se hamesha return karo. console.log sirf test karne ke liye hai.
- Code ho jaye ya kahin atko to apna code Claude ko bhejo, wo hint ke saath check karega.
- Din mein 2-3 problems kaafi hain. Jaldi mat karo, har problem ka dry run zaroor karo. Set 1 pura karne ke baad ye set shuru karo.

Code bhejne se pehle checklist (aapki ab tak ki galtiyon se)

[ ] Loop: `i < arr.length` hai, <= to nahi? (warna aakhir mein undefined / NaN aata hai)

[ ] Comparison mein === hai, akela = to nahi? (`if (x = 0)` value rakhta hai, compare nahi karta)

[ ] `return` sahi jagah hai? Jawab sab elements dekhne ke baad chahiye to loop ke bahar, pehla match milte hi chahiye to loop ke andar.

[ ] Function return karta hai ya sirf print? Print karne se bahar wale console.log mein undefined aata hai.

[ ] Counter / total ki shuruaati value sahi hai? (sum ke liye 0, multiplication ke liye 1, khali array par kya hoga?)

Problem 11: Product of an Array

Level: Easy

New concept: Multiplication accumulator: shuruaati value 0 nahi, 1 hoti hai.

DESCRIPTION

Given an array of numbers, return the **product** of all its elements (multiply them together).

INPUT / OUTPUT

Input: An array of numbers, arr.

Output: One number: the product of all elements. Return 1 for an empty array.

EXAMPLES

Input	Output	Note
`[1, 2, 3, 4]`	`24`	1 x 2 x 3 x 4
`[5]`	`5`	One element
`[2, 0, 3]`	`0`	Anything times 0 is 0
`[-2, 3]`	`-6`	Negatives work too
`[]`	`1`	Empty product is 1

CONSTRAINTS

- Use a for loop.
- Do not use reduce().
- Return the product, do not just print it.

STARTER CODE AND TEST CASES

```
function productArray(arr) {
  // your code here
}

console.log(productArray([1, 2, 3, 4]));  // 24
console.log(productArray([5]));           // 5
console.log(productArray([2, 0, 3]));     // 0
console.log(productArray([-2, 3]));       // -6
console.log(productArray([]));            // 1
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Ye array ke sum wale problem jaisa hai, bas jodne ki jagah multiply karna hai.

Hint 2: Sum mein shuruaat 0 se hoti thi. Agar yahan bhi 0 se shuru karo to har jawab kya aayega? Factorial wala problem yaad karo.

Hint 3: Shuruaati value 1 rakho, to khali array par loop chalta hi nahi aur apne aap 1 return hota hai.

Problem 12: Reverse an Array

Level: Easy

New concept: Array ko peeche se parhna aur naya array banana (original ko na badalna).

DESCRIPTION

Given an array, return a **new array** with the elements in reverse order.

INPUT / OUTPUT

Input: An array, arr.

Output: A new array with the same elements in reverse order.

EXAMPLES

Input	Output	Note
`[1, 2, 3, 4]`	`[4, 3, 2, 1]`	
`["a", "b", "c"]`	`["c", "b", "a"]`	Works with strings too
`[7]`	`[7]`	One element
`[1, 2, 2, 3]`	`[3, 2, 2, 1]`	Duplicates are fine
`[]`	`[]`	Empty array

CONSTRAINTS

- Use a for loop.
- Do not use reverse().
- Do not change the original array. Return a new one.

STARTER CODE AND TEST CASES

```
function reverseArray(arr) {
  // your code here
}

console.log(reverseArray([1, 2, 3, 4]));      // [4, 3, 2, 1]
console.log(reverseArray(["a", "b", "c"]));  // ["c", "b", "a"]
console.log(reverseArray([7]));               // [7]
console.log(reverseArray([]));                // []
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Reverse string wala problem yaad karo: loop aakhri index se 0 tak ulta chalta tha.

Hint 2: Ek khali array banao (let result = []) aur ulte loop mein har element ko result.push(...) se daalte jao.

Hint 3: Aakhri index arr.length - 1 hai. Loop ki condition i >= 0 hogi. return loop ke bahar likho.

Problem 13: Capitalize the First Letter

Level: Easy

New concept: String ko loop se dobara banana, aur khali string ka edge case.

DESCRIPTION

Given a word, return it with the **first letter in uppercase**. The rest of the letters stay as they are.

INPUT / OUTPUT

Input: A string, word.

Output: A new string with the first letter capitalized.

EXAMPLES

Input	Output	Note
`"hello"`	`"Hello"`	
`"pakistan"`	`"Pakistan"`	
`"a"`	`"A"`	One letter
`"javaScript"`	`"JavaScript"`	Rest stays unchanged
`""`	`""`	Empty string

CONSTRAINTS

- Use a for loop to build the rest of the word.
- Do not use slice(), substring() or replace().
- toUpperCase() is allowed.

STARTER CODE AND TEST CASES

```
function capitalizeFirst(word) {
  // your code here
}

console.log(capitalizeFirst("hello"));       // "Hello"
console.log(capitalizeFirst("pakistan"));    // "Pakistan"
console.log(capitalizeFirst("a"));           // "A"
console.log(capitalizeFirst("javaScript"));  // "JavaScript"
console.log(capitalizeFirst(""));            // ""
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Pehla letter alag handle karo: word[0].toUpperCase(). Baqi letters jaise hain waise hi rakhne hain.

Hint 2: result ko pehle letter (capital) se shuru karo, phir loop i = 1 se chalao aur har letter result mein jodte jao.

Hint 3: Khali string par word[0] undefined hota hai aur undefined.toUpperCase() crash karta hai. Function ke shuru mein is case ko pehle sambhalo.

Problem 14: Count Capital Letters

Level: Easy+

New concept: Character ko category ke hisab se pehchanna (capital, small, digit, space).

DESCRIPTION

Given a string, return **how many capital (uppercase) letters** it contains. Digits, spaces and small letters do not count.

INPUT / OUTPUT

Input: A string, str.

Output: One number: the count of uppercase letters.

EXAMPLES

Input	Output	Note
`"Hello World"`	`2`	H and W
`"abc"`	`0`	No capitals
`"ABC"`	`3`	All capitals
`"a1B2"`	`1`	Only B. Digits do not count
`""`	`0`	Empty string

CONSTRAINTS

- Use a for loop.
- Do not use match() or regular expressions.
- Return the count, do not just print it.

STARTER CODE AND TEST CASES

```
function countCapitals(str) {
  // your code here
}

console.log(countCapitals("Hello World"));  // 2
console.log(countCapitals("abc"));          // 0
console.log(countCapitals("ABC"));          // 3
console.log(countCapitals("a1B2"));         // 1
console.log(countCapitals(""));             // 0
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Ye counter wala pattern hai, bilkul vowels ginne jaisa. Bas shart badalni hai.

Hint 2: JavaScript mein letters ko alphabet ke order se compare kar sakte ho: ch >= "A" && ch <= "Z" sirf capital letters par true hota hai.

Hint 3: Dusra tareeqa: ch === ch.toUpperCase() akela kaafi nahi, kyunke digit aur space bhi apne uppercase ke barabar hote hain. Is ke saath ek aur condition lagani parti hai. Kaunsi?

Problem 15: Count Words in a Sentence

Level: Easy+

New concept: Characters dekh kar structure pehchanna (space = word ki hadd).

DESCRIPTION

Given a sentence, return **how many words** it has. Words are separated by a single space, and there are no extra spaces at the start or end.

INPUT / OUTPUT

Input: A string, sentence.
Output: One number: the number of words.

EXAMPLES

Input	Output	Note
`"I love JavaScript"`	`3`	
`"hello"`	`1`	One word, no spaces
`"one two"`	`2`	
`"a b c d e"`	`5`	
`""`	`0`	Empty string has no words

CONSTRAINTS

- Use a for loop.
- Do not use split().
- Return the count, do not just print it.

STARTER CODE AND TEST CASES

```
function countWords(sentence) {
  // your code here
}

console.log(countWords("I love JavaScript"));  // 3
console.log(countWords("hello"));              // 1
console.log(countWords("one two"));            // 2
console.log(countWords("a b c d e"));          // 5
console.log(countWords(""));                   // 0
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: 3 words ke beech kitne spaces hote hain? 2. To words ka kya taluq hai spaces ki tadaad se?

Hint 2: Loop se spaces gino (sentence[i] === " "). Phir words = spaces + 1.

Hint 3: Khali string par spaces 0 hain, lekin words 0 hone chahiye, 1 nahi. Is special case ko pehle hi sambhal lo.

Problem 16: Power of a Number

Level: Easy+

New concept: Repeated multiplication: loop ko 'kitni baar' ke hisab se chalana.

DESCRIPTION

Given a **base** and an **exponent**, return base raised to the power of exponent (for example 2 to the power 3 is 2 x 2 x 2 = 8).

INPUT / OUTPUT

Input: Two numbers: base and exp (exp is a whole number, 0 or greater).
Output: One number: base to the power exp.

EXAMPLES

Input	Output	Note
`2, 3`	`8`	2 x 2 x 2
`5, 0`	`1`	Anything to the power 0 is 1
`3, 1`	`3`	
`10, 4`	`10000`	
`0, 5`	`0`	

CONSTRAINTS

- Use a for loop.
- Do not use ** or Math.pow().
- Return the result, do not just print it.

STARTER CODE AND TEST CASES

```
function power(base, exp) {
  // your code here
}

console.log(power(2, 3));   // 8
console.log(power(5, 0));   // 1
console.log(power(3, 1));   // 3
console.log(power(10, 4));  // 10000
console.log(power(0, 5));   // 0
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: 2 ka power 3 matlab 2 ko 3 baar apne se multiply karna. Ye factorial jaisa accumulator hai.

Hint 2: result = 1 se shuru karo, aur loop ko exp baar chalao. Har baar result ko base se multiply karo.

Hint 3: exp = 0 par loop chalta hi nahi, aur result = 1 hi return hota hai, jo sahi jawab hai.

Problem 17: Find Min and Max Together

Level: Medium-Easy

New concept: Ek hi loop mein do cheezein track karna, aur object return karna.

DESCRIPTION

Given an array of numbers, find both the **smallest** and the **largest** number in a **single loop**. Return them in an object. Return **null** for an empty array.

INPUT / OUTPUT

Input: An array of numbers, arr.
Output: An object { min: ?, max: ? }, or null.

EXAMPLES

Input	Output	Note
`[3, 7, 2, 9, 4]`	`{ min: 2, max: 9 }`	
`[5]`	`{ min: 5, max: 5 }`	One element is both
`[-3, -1, -8]`	`{ min: -8, max: -1 }`	All negative
`[4, 4]`	`{ min: 4, max: 4 }`	Duplicates
`[]`	`null`	Empty array

CONSTRAINTS

- Use only one for loop (not two).
- Do not use Math.min(), Math.max() or sort().
- Return an object with the keys min and max.

STARTER CODE AND TEST CASES

```
function findMinMax(arr) {
  // your code here
}

console.log(findMinMax([3, 7, 2, 9, 4]));  // { min: 2, max: 9 }
console.log(findMinMax([5]));              // { min: 5, max: 5 }
console.log(findMinMax([-3, -1, -8]));     // { min: -8, max: -1 }
console.log(findMinMax([]));               // null
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Aap sabse bada aur sabse chhota dono alag problems mein kar chuke ho. Ab dono ko ek loop mein jodo.

Hint 2: Do variables rakho (min aur max), dono ko arr[0] se shuru karo, 0 se nahi. Kyun? Negative numbers par 0 galat jawab deta hai.

Hint 3: Loop ke andar do alag if likho: ek min ke liye, ek max ke liye. Khali array ka check function ke shuru mein karo.

Problem 18: Prime Number Check

Level: Medium-Easy

New concept: Early return aur edge cases: 0, 1, 2 ke liye khaas sochna.

DESCRIPTION

A **prime number** is greater than 1 and can only be divided evenly by 1 and itself. Given a number, return **true** if it is prime, otherwise **false**.

INPUT / OUTPUT

Input: A whole number n (0 or greater).
Output: true or false.

EXAMPLES

Input	Output	Note
`7`	`true`	Only divisible by 1 and 7
`2`	`true`	Smallest prime
`1`	`false`	1 is not prime
`0`	`false`	
`9`	`false`	3 x 3
`20`	`false`	Divisible by 2, 4, 5, 10

CONSTRAINTS

- Use a for loop.
- Return a boolean (true / false), not a string.
- Stop checking as soon as you find a divisor.

STARTER CODE AND TEST CASES

```
function isPrime(n) {
  // your code here
}

console.log(isPrime(7));    // true
console.log(isPrime(2));    // true
console.log(isPrime(1));    // false
console.log(isPrime(0));    // false
console.log(isPrime(9));    // false
console.log(isPrime(20));   // false
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Pehle chhote numbers dekho: 0 aur 1 prime nahi hote. Is liye function ke shuru mein n < 2 par kya return hoga?

Hint 2: n ko 2 se n - 1 tak ke har number se divide karke dekho. Agar koi number n ko poora divide kar de (n % i === 0), to n prime nahi hai.

Hint 3: Jaise hi divisor mile, wahin return false (loop ke andar). return true loop ke bahar likho. n = 2 par loop chalta hi nahi aur true aata hai, jo sahi hai.

Problem 19: Fibonacci Series

Level: Medium-Easy

New concept: Pichli values se agli value banana (array mein pichle do elements dekhna).

DESCRIPTION

In the Fibonacci series each number is the **sum of the two numbers before it**, starting with 0 and 1: 0, 1, 1, 2, 3, 5, 8, ... Given n, return an array with the **first n** Fibonacci numbers.

INPUT / OUTPUT

Input: A non-negative integer n.
Output: An array of length n.

EXAMPLES

Input	Output	Note
`7`	`[0, 1, 1, 2, 3, 5, 8]`	
`10`	`[0, 1, 1, 2, 3, 5, 8, 13, 21, 34]`	
`2`	`[0, 1]`	Just the starting pair
`1`	`[0]`	
`0`	`[]`	Nothing to build

CONSTRAINTS

- Use a for loop.
- Do not use recursion.
- Return the array, do not just print it.

STARTER CODE AND TEST CASES

```
function fibonacci(n) {
  // your code here
}

console.log(fibonacci(7));   // [0, 1, 1, 2, 3, 5, 8]
console.log(fibonacci(10));  // [0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
console.log(fibonacci(2));   // [0, 1]
console.log(fibonacci(1));   // [0]
console.log(fibonacci(0));   // []
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Pehle do numbers (0 aur 1) fixed hain. Teesra number inka sum hai (0 + 1 = 1), chautha pichle do ka sum (1 + 1 = 2), aur isi tarah.

Hint 2: Ek array banao aur usme pehle 0 aur 1 daalo. Phir loop i = 2 se n tak chalao aur result[i - 1] + result[i - 2] push karte jao.

Hint 3: n = 0, 1 aur 2 par special dhyan do. Pehle in teeno cases ko if se sambhal lo aur seedha return kar do, phir baqi n ke liye loop chalao.

Problem 20: Has Duplicate (Nested Loops)

Level: Medium

New concept: Nested loop: loop ke andar loop. Har element ko baqi sab se compare karna.

DESCRIPTION

Given an array, return **true** if any value appears **more than once**, otherwise return **false**.

INPUT / OUTPUT

Input: An array of numbers, arr.

Output: true or false.

EXAMPLES

Input	Output	Note
`[1, 2, 3, 1]`	`true`	1 appears twice
`[1, 2, 3]`	`false`	All different
`[5, 5]`	`true`	
`[7]`	`false`	One element cannot repeat
`[]`	`false`	Empty array

CONSTRAINTS

- Use two for loops (one inside the other).
- Do not use Set, indexOf() or includes().
- Return a boolean.

STARTER CODE AND TEST CASES

```
function hasDuplicate(arr) {
  // your code here
}

console.log(hasDuplicate([1, 2, 3, 1]));  // true
console.log(hasDuplicate([1, 2, 3]));     // false
console.log(hasDuplicate([5, 5]));        // true
console.log(hasDuplicate([7]));           // false
console.log(hasDuplicate([]));            // false
```

Hints (sirf atko to dekhna, ek ek kar ke)

Hint 1: Pehle element ko baqi sab elements se compare karo, phir doosre ko, aur isi tarah. Ye 'ek ke andar ek' loop hai.

Hint 2: Bahar wala loop i se, andar wala j se chalega. j ko i + 1 se shuru karo. Agar j = 0 se shuru karoge to element khud se compare hoga aur har array par true aayega.

Hint 3: Jaise hi arr[i] === arr[j] mile, wahin return true. return false dono loops ke baad likho. Copy par [1, 2, 3, 1] ka dry run karo, table mein i aur j dono rakho.

Progress Tracker

Har problem ke saamne jo step mukammal ho, wahan tick (X) lagao.

#	Problem	Algorithm	Pseudocode	Dry run	Code	All tests pass
11	Product of an Array					
12	Reverse an Array					
13	Capitalize the First Letter					
14	Count Capital Letters					
15	Count Words in a Sentence					
16	Power of a Number					
17	Find Min and Max Together					
18	Prime Number Check					
19	Fibonacci Series					
20	Has Duplicate (Nested Loops)					

Bonus sawal (har problem ke baad khud se poocho)

- Is code ki time complexity kya hai? Kitni baar loop chalta hai? (Nested loop ho to kitni baar?)
- Is code ki space complexity kya hai? Kya naya array ya string bana?
- Best case aur worst case mein kya farq hai?
- Kya main isi problem ko doosre tareeqe se bhi hal kar sakta hoon?